EE 354/CE 361/MATH 310 -

Introduction to Probability

and Statistics

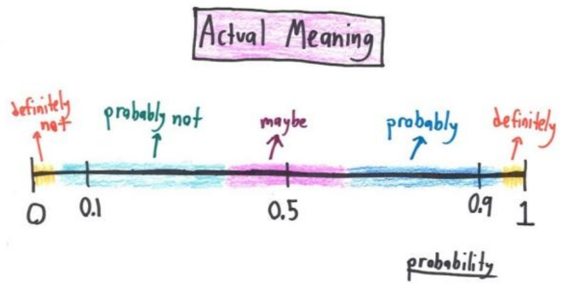
Spring 2026

EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

Aamir Hasan

Week 10 - Lecture No 19 – March 16, 2026



Announcements!

1. Office hours (Ramadan) – Tuesday & Thursday 11:30 to 12:30 PM
2. Quiz No 05 – March 18, 2026
    1. Quizzes – rule for N-1 would be instituted after grading of Quiz No 5 (please note your own standing in the course)
3. Online class on March 25, 2026 (accommodation is requested)
    1. Cameras to remain switched on during the entire lecture
    2. Please ensure audio and video is working (see below)
    3. I will place questions to the students and not responding w/ audio & video would be used for participation & attendance. In cases where I am unable to ask questions to individuals, attendance would be taken sometime in between the lecture.
    4. Slides for the online class will be posted (after some delay)
4. MT exam showing (tomorrow, 11 – 12:30) – last chance
5. TA's EHSAS Hours

EHSAS TUTORS' RAMADAN SCHEDULE - Spring 2026

Course	Tutor	Email	Monday	Tuesday	Wednesday	Thursday	Friday
Intro to Probability & Statistics	Rayyan Ali Shah	rs08770@st.habib.edu.pk	2:00-3:00		2:00-3:00		12:40-1:40
	Hiba Aiman Ali	ha09269@st.habib.edu.pk	12:45-2:30		12:45-2:30		11:00-12:00
	Zain Naqi	zn09224@st.habib.edu.pk		9:00 - 10:30	11:45 - 12:45	9:00 - 10:30	

Agenda for today

1. Unit 4 – Discrete Random Variable (to be completed)

Independence

- Of two events: $\mathbf{P}(A \cap B) = \mathrm{P}(A)\,\mathrm{P}(B)$ $\qquad \mathbf{P}(A \mid B) = \mathrm{P}(A)$
- of a r.v. and an event: $\mathrm{P}(X = x \text{ and } \mathrm{A}) = \mathrm{P}(X = x)\,\mathrm{P}(\mathrm{A})$ *for all x*

$\Rightarrow p_{X|A}(x) = p_X(x)$ *for all x* $\quad \Rightarrow P(A \mid X = x) = P(A)$ *for all x*

- of two r.v.'s: $\mathrm{P}(X = x \text{ and } \mathrm{Y} = \mathrm{y}) = \mathrm{P}(X = x)\,\mathrm{P}(\mathrm{Y} = \mathrm{y})$ *for all x, y*

$\Rightarrow p_{X|Y}(x|y) = p_X(x)$

$\Rightarrow p_{Y|X}(y|x) = p_Y(y)$

$$p_{X,Y}(x, y) = p_X(x)\, p_Y(y)$$

X, Y, Z are **independent** if:
$p_{X,Y,Z}(x, y, z) = p_X(x)\, p_Y(y)\, p_Z(z),$ *for all x, y, z*

Example: independence and conditional independence

Y = \ X =	1	2	3	4
4	1/20	2/20	2/20	0
3	2/20	4/20	1/20	2/20
2	0	1/20	3/20	1/20
1	0	1/20	0	0

- Independent? No

➡ $P_X(1) = 3/20$

➡ $P_{X|Y}(1|1) = 0$

- What if we condition on A ($X \leq 2$ and $Y \geq 3$)? Yes

➡ $P_{X|A}(1) = 1/3$, $P_{X|A}(2) = 2/3$

➡ $P_{Y|A}(3) = 2/3$, $P_{Y|A}(4) = 1/3$

➡ $P_{X|Y}(1|3) = 1/3$

	X = 1	X = 2	Y =
1/3	1/9	2/9	4
2/3	2/9	4/9	3
	1/3	2/3	

Conditioned on $X \leq 2$ and $Y \geq 3$

Conditioned on A i.e. ($X \leq 2$ and $Y \geq 3$)

$P_{X,Y|A}(x,y) = P_{X|A}(x).P_{Y|A}(y)$ for x (= 1, 2), y (= 3, 4)

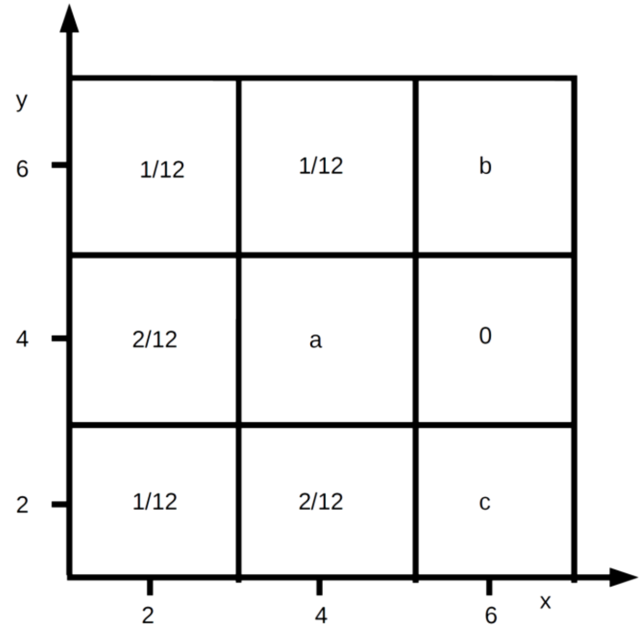
Exercise No 2

Sample Space for two events x and y with probabilities of event $x \bigcap y$ are given in Fig. 1

(a) (**3 marks**) Calculate the probability that $x = 2$.

(b) (**3 marks**) Find and sketch the probabilities $P_{y|x}(y|x = 2)$.

(c) (**4 marks**) Find the probabilities (a, b, c) under which event x and y are independent each other. You need to provide proper justification and working for your answer.



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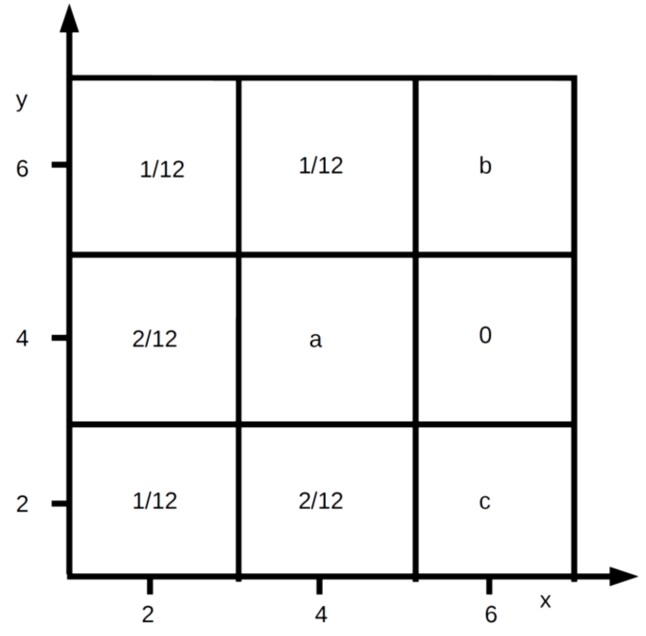

(a) $P(x=2) = \sum_{y,\, x=2} P(x, y)$

$= P(2,2) + P(2,4) + P(2,6)$

$= 1/12 + 2/12 + 1/12 = 4/12$

$P(x=2) = 1/3$

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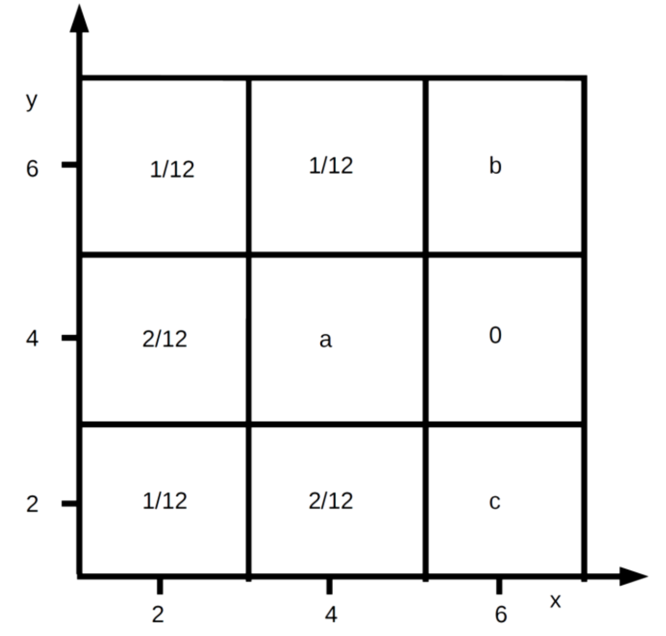

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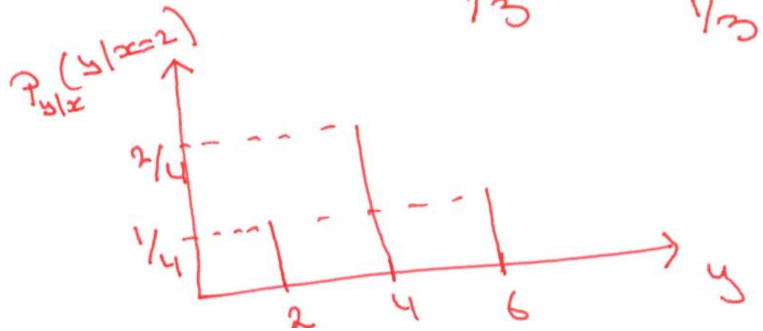
(b) $P(y|x=2) = \dfrac{P(y \cap (x=2))}{P(x=2) = 1/3}$

$y=2 \longrightarrow = \dfrac{P(2,2)}{1/3} = \dfrac{1/12}{1/3} = 1/4$

$y=4 \longrightarrow = \dfrac{P(2,4)}{1/3} = \dfrac{2/12}{1/3} = \dfrac{2}{4} = 1/2$

$y=6 \longrightarrow = \dfrac{P(2,6)}{1/3} = \dfrac{1/12}{1/3} = 1/4$



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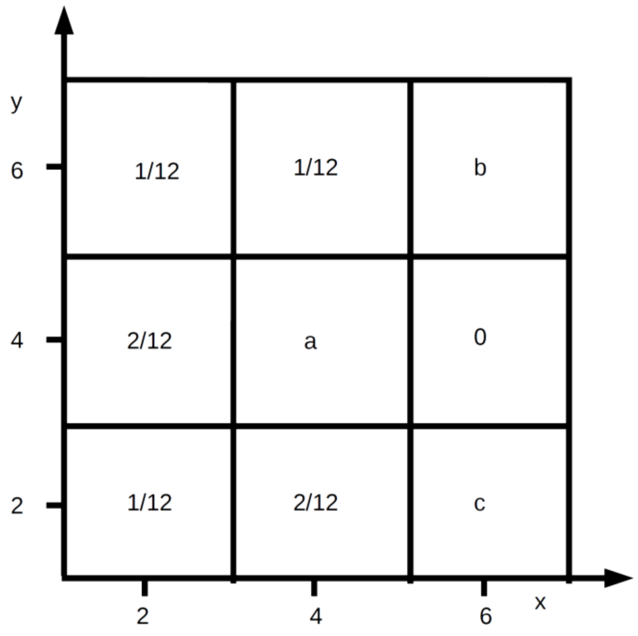


c) When $x = 6$, y cannot be 4

$\Rightarrow P(y=4|x=6) = 0$

however $P(y=4|x=2) = 2/4 \neq 0$

This implies that x, y can never be independent and hence there exists no value for a, b, c that makes x, y Independent.

Independence and Expectations

- In general: $\mathbf{E}[\, g(X,Y)\,] \neq g(\,\mathbf{E}[X], \mathbf{E}[Y]\,)$ always true
- Exceptions: $\mathbf{E}[aX+b] = a\mathrm{E}[X] + b$ $\quad$ $\mathbf{E}[X+Y+Z] = \mathbf{E}[X] + \mathbf{E}[Y] + \mathbf{E}[Z]$

If X, Y are **independent**: $\mathbf{E}[XY] = \mathbf{E}[X]\,\mathbf{E}[Y]$

g(X) and h(Y) are also independent: $\mathbf{E}[g(X)\,h(Y)] = \mathbf{E}[g(X)]\,\mathbf{E}[h(Y)]$

$\mathbf{E}[g(X,Y)]$ $\qquad$ $g(x,y) = xy$

$$\Rightarrow \quad = \sum_x \sum_y (xy)\, p_{X,Y}(x,y) = \sum_x \sum_y (xy)\, p_X(x)\, p_Y(y)$$

$$\Rightarrow \quad = \sum_x x\, p_X(x) \sum_y y\, p_Y(y) = E[X]\; E[Y]$$

Independence and Variances

- Always true: $var(aX + b) = a^2 var(X)$ $\quad var(X + a) = var(X)$

- In general: $var(X + Y) \neq var(X) + var(Y)$

If X, Y are **independent**: $var(X + Y) = var(X) + var(Y)$

assume

$E[X] = 0$; $E[Y]=0$

$E[X]E[Y] = 0$

$\Rightarrow$ $var(X+Y) = E\left[(X+Y)^2\right] = E[(X^2 + 2XY + Y^2)]$

$\Rightarrow$ $= E[X^2] + 2E[XY] + E[Y^2] = E[X^2] + E[Y^2] = var(X) + var(Y)$

- Examples:
  - If X = Y: var(X + Y) $= var(2X) = 4var(X)$
  - If X = -Y: var(X + Y) $= var(0) = 0$
  - If X , Y independent: var(X - 3Y) $= var(X) + var(-3Y) = var(X) + 9var(Y)$

Poisson RV

Parameter: λ (a positive real number characterizing the average number of events in the interval)

$$P_X(k) = e^{-\lambda}\frac{\lambda^K}{K!}, K = 0, 1, 2, 3, . \ . \ .$$

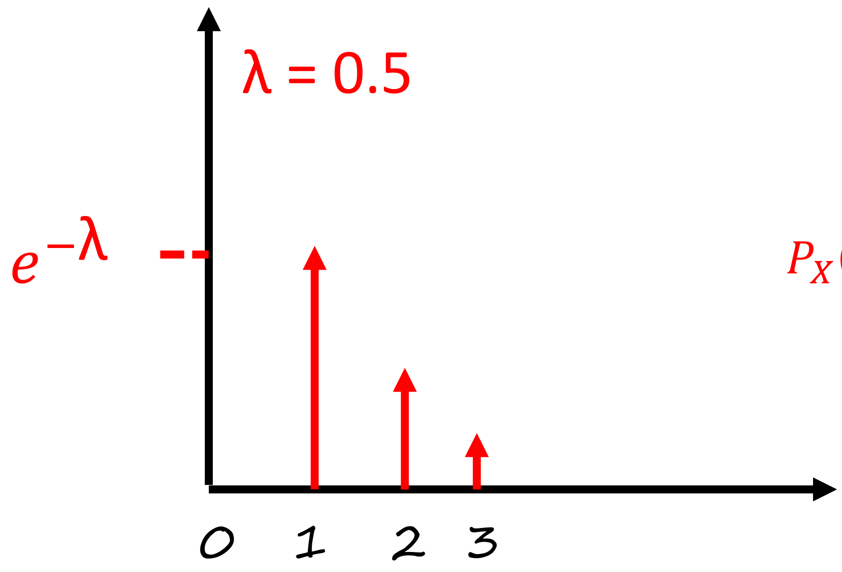
Models: Probability of occurrence of events in a fixed interval of time or space if these events occur with a known constant rate and independent of time of last event.

Note: $\sum_{K=0}^{\infty} e^{-\lambda}\frac{\lambda^K}{K!} = e^{-\lambda}(1 + \lambda + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} \dots)$

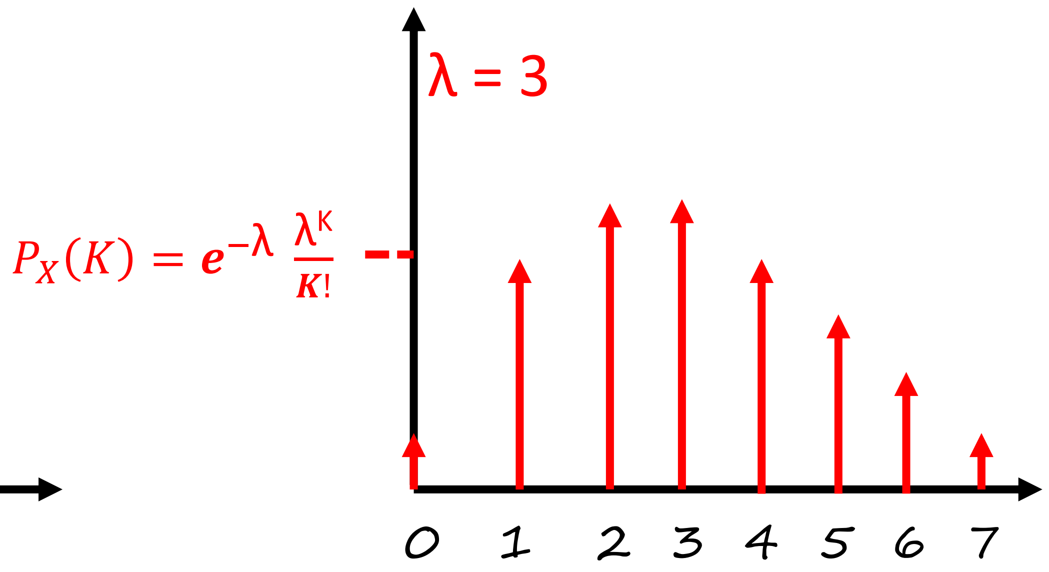
$= e^{-\lambda} * e^{\lambda} = 1$

Recognize the Taylor series: The remaining sum, $\sum_{k=0}^{\infty}\frac{\lambda^k}{k!}$, is the well-known Taylor series expansion of e^x evaluated at $x = \lambda$. The general form is:

$$e^x = \sum_{k=0}^{\infty}\frac{x^k}{k!} = \frac{x^0}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \dots$$



Note: When λ < 1, PMF, is monotonically decreasing with K



Note: When λ > 1, PMF first increases and then decreases

Poisson RV: Example

$$\boldsymbol{P_X(k) = e^{-\lambda}\,\frac{\lambda^{K}}{K!}}$$

An IC manufacturing company provides 100 IC's per day. Defective IC's are at a fixed rate of 3 IC's/day. Probability of R defective IC's in a given day.

$$P_X(0) = e^{-3} \approx 0.05$$

$$P_X(3) = e^{-3}.\frac{3^3}{3!} = 0.05 \text{ x } 4.5 = 0.2256$$

How do you find P(X > 3) ?

Useful in Modeling

a) No. of phone calls received by a call center (per hour, per day, or per anytime unit)
b) No. of photons hitting a detector in a particular time interval

Variance of the binomial

- *X*: binomial with parameters *n*, *p*
  - ❖ number of successes in n independent trials

$X_i = 1$ if i^{th} trial is a success; p

(indicator variable)

$X_i = 0$ otherwise $1-p$

independent

$$\boldsymbol{X} = \boldsymbol{X_1} + \ldots + \boldsymbol{X_n}$$

$\Rightarrow$ $var\ (X) = var\ (X_1) + var\ (X_2) + \ldots + var\ (X_n)$

$\Rightarrow$ $var\ (X) = nvar\ (X_1) = \boxed{np(1-p)}$

The variance of the geometric PMF

X: geometric parameter with p

- Given: X > 1, X – 1 has the same geometric PMF (unconditional PMF of X)
- Given X > 1, conditional PMF of X: same as unconditional PMF of X + 1

➡ $E[X \mid X>1] = 1 + E[X]$ $\quad E[X^2 \mid X>1] = E[(X+1)^2]$

➡ $E[X^2] = P(X=1)\,E[X^2 \mid X=1] + P(X>1)\,E[X^2 \mid X>1]$

$= p*1 + (1-p)\,(E[X^2] + 2E[X] + 1)$

1/p

Algebra: $E[X^2] = \frac{2}{p^2} - \frac{1}{p}$

$Var(x) = E[X^2] - (E[X])^2$

$$\boxed{Var(X) = \frac{1-p}{p^2}}$$

Discrete Random Variables - Summary

- r.v.'s and PMFs: $p_X(x), p_{X,Y}, p_{X|Y}(x|y), p_{X|A}(x)$
- Expectation: $\mathrm{E[X], E[X|A], E[X|Y = y]}$
  Expected value rule: E[g(X,Y)], E[g(X,Y)|A], E[g(X,Y) | Z = z]
  Linearity: E[aX + bY] = aE[Y] + bE[Y]
- Variance: var(X), var(X|A), var(X|Y = y) $\quad var(X) = E[X^2] - (E[X])^2$
- Independence of r.v.'s: $p_{X,Y} = p_X * p_Y$
  E[XY] = E[X] * E[Y] $\quad$ var(X + Y) = var(X) + var (Y)
- Multiplication Rule $\quad p_{X,Y,Z}(x,y,z) = p_Z(z)\, p_{Y|Z}(y|z)\, p_{X|Y,Z}(x|y,z)$
- Total Probability Theorem $\quad p_X(x) = \sum_y p_Y(y) p_{X|Y}(x|y)$
- Total expectation Theorem $\quad$ E[X] = $\sum_y p_Y(y) E[X|Y = y]$
- Examples: Bernoulli, indicators, uniform, binomial, geometric

End of Unit 4!

Practice Problems

Unit 4!

Exercise No 1

A Random Variable Z_n is defined as $Z_n = min(X_1; X_2; \dots ; X_n)$, where $X_i's$ are outcomes of identical and independently distributed fair six-sided die. As an example $Z_2 = min(X_1; X_2)$ - meaning Z_2 is the minimum of the outcome of two independent throws of six-sided die X_1 and X_2.

(a) (**5 marks**) Calculate probability $P(1 < Z_{10} < 6)$.

(b) (**5 marks**) Find the minimum value of n such that $P(Z_n = 1) \geq 0.6$

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(a) (**5 marks**) Calculate probability $P(1 < Z_{10} < 6)$.

(a) $P(1 < Z_{10} < 6) = 1 - P(Z_{10} = 1) - P(Z_{10} = 6)$

$P(Z_{10} = 1) = 1 - P(Z_{10} \neq 1)$

$= 1 - P(Z_{10} \geq 2)$

$= 1 - P($All 10 die are between 2−6$)$

$P(Z_{10} = 1) = 1 - \left(\frac{5}{6}\right)^{10}$

$P(Z_{10} = 6) = \cancel{1}\, P($All 10 die are 6$)$

$= \left(\frac{1}{6}\right)^{10}$

$P(Z_{10} = 6) = \cancel{1}\, P($All 10 die are 6$)$

$= \left(\frac{1}{6}\right)^{10}$

$P(1 < Z_{10} < 6) = 1 - \left(1 - \left(\frac{5}{6}\right)^{10}\right) - \left(\frac{1}{6}\right)^{10}$

$= 1 - 1 + \left(\frac{5}{6}\right)^{10} - \left(\frac{1}{6}\right)^{10}$

$= \left(\frac{5}{6}\right)^{10} - \left(\frac{1}{6}\right)^{10}$ 0.1615

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(b) (**5 marks**) Find the minimum value of n such that $P(Z_n = 1) \geq 0.6$

$P(Z_n = 1) = 1 - P(Z_n \neq 1)$

$= 1 - P(\text{All } n \text{ throws are greater than } 1)$

$= 1 - \left(\frac{5}{6}\right)^n$

$\Rightarrow 1 - (5/6)^n \geq 0.6$

$\left(\frac{5}{6}\right)^n \leq 0.4 \Rightarrow n \log 5/6 \leq \log 0.4$

$n \geq \frac{-0.397}{-0.079}$

$n \geq 5.02$

$n = 6$

Question 1 - (15 points)

Let's be optimistic and assume that Pakistan cricket team has qualified for the T20 World Cup Semi Final. Pakistan's opponent in Semi Final will be West Indies, England, or Australia with a probability of 0.4, 0.4, and 0.2 respectively. The probability of Pakistan winning against West Indies is 0.6. The probability of Pakistan winning against England is 0.4. The probability of Pakistan winning against Australia is 0.6.

a) (7.5 points) What is the probability that Pakistan will win the Semi-final.

b) (7.5 points) Assume that you decided to completely switch off from cricket and did not follow the game. Later, you were told that Pakistan lost the Semi Final. What is the probability that Pakistan played the Semi Final game against England?

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Solution:

(a)

Let's assume the following naming conventions for various events:

$A_1 = \{$Pakistan play West Indies in Semi Final$\}$

$A_2 = \{$Pakistan play England in Semi Final$\}$

$A_3 = \{$Pakistan play Australia in Semi Final$\}$

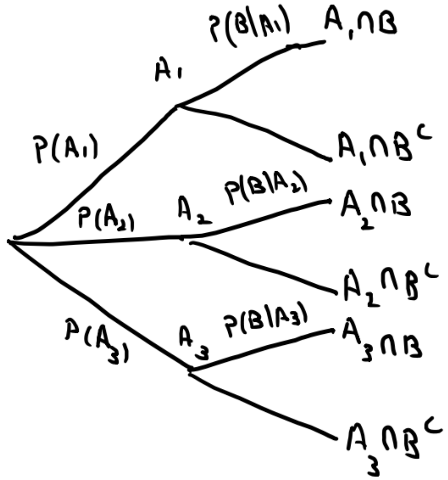
$B = \{$Pakistan win the Semi Final$\}$

$B^c = \{$Pakistan lose the Semi Final$\}$

$\Rightarrow$ $P(A_1) = 0.4$ $\quad P(A_2) = 0.4$ $\quad P(A_3) = 0.2$

$P(B|A_1) = 0.6$ $\quad P(B|A_2) = 0.4$ $\quad P(B|A_3) = 0.6$

$P(B^c|A_1) = 0.4$ $\quad P(B^c|A_2) = 0.6$ $\quad P(B^c|A_3) = 0.4$



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$\Rightarrow$ $P(A_1) = 0.4$ $P(A_2) = 0.4$ $P(A_3) = 0.2$

$P(B|A_1) = 0.6$ $P(B|A_2) = 0.4$ $P(B|A_3) = 0.6$

$P(B^c|A_1) = 0.4$ $P(B^c|A_2) = 0.6$ $P(B^c|A_3) = 0.4$

By Total Probability Theorem:-

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B)$$

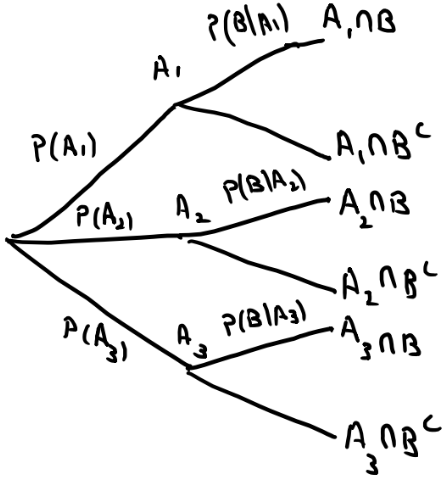
$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)$$

$$= (0.4)(0.6) + (0.4)(0.4) + (0.2)(0.6)$$

$$= 0.24 + 0.16 + 0.12$$

$$P(B) = 0.52 \quad \text{And}$$

$$P(B^c) = 1 - P(B) = 0.48$$



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$P(B^c) = 1 - P(B) = 0.48$

According to Bayes' Rule,

$$P(A_2|B^c) = \frac{P(A_2)P(B^c|A_2)}{P(B^c)}$$

$$= \frac{(0.4)(0.6)}{0.48}$$

$P(A_2|B^c) = 0.5$ — Ans (b)

$\Rightarrow$ $P(A_1) = 0.4$ $P(A_2) = 0.4$ $P(A_3) = 0.2$

$P(B|A_1) = 0.6$ $P(B|A_2) = 0.4$ $P(B|A_3) = 0.6$

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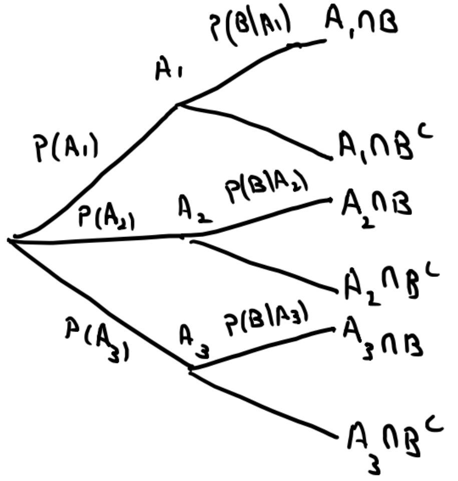

Question 3 - (14 points)

Consider the following random variables X & Y:

X = Magnitude of the difference between the outcomes of two rolls of a fair 3-sided die.

(If the first roll results in 1 and second roll results in 3, the value taken by X is 2)

(If the first roll results in 3 and second roll results in 1, the value taken by X is 2)

Y = 2X

a) (4 points) Calculate and plot the PMF of X

b) (4 points) Calculate and plot the PMF of Y

c) (4 points) Calculate Mean of Y.

d) (4 points) Calculate Variance of Y

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b) (4 points) Calculate and plot the PMF of Y

c) (4 points) Calculate Mean of Y.

d) (4 points) Calculate Variance of Y

Possible magnitudes of the difference between two rolls:

Roll 2 = 3	2	1	0
Roll 2 = 2	1	0	1
Roll 2 = 1	0	1	2
	Roll 1 = 1	Roll 1 = 2	Roll 1 = 3

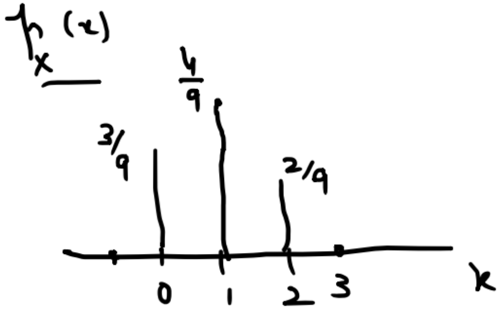
(a)

Possible values that X can take = $\{0, 1, 2\}$

$P(X=0) = \frac{3}{9}$

$P(X=1) = \frac{4}{9}$

$P(X=2) = \frac{2}{9}$



Question 3 - (14 points)

Consider the following random variables X & Y:

X = Magnitude of the difference between the outcomes of two rolls of a fair 3-sided die.

(If the first roll results in 1 and second roll results in 3, the value taken by X is 2)

(If the first roll results in 3 and second roll results in 1, the value taken by X is 2)

Y = 2X

a) (4 points) Calculate and plot the PMF of X

b) (4 points) Calculate and plot the PMF of Y

c) (4 points) Calculate Mean of Y.

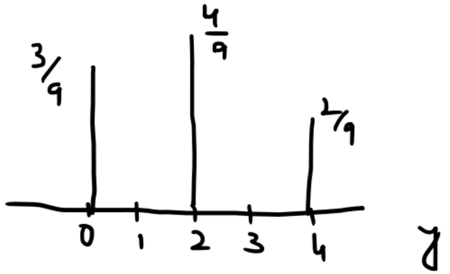
d) (4 points) Calculate Variance of Y

(b)

$Y = 2X$

Possible values of $Y = \{0, 2, 4\}$

$p_Y(y)$:



(c)

$$\mu_Y = E[Y] = \sum_y y\, p_Y(y)$$

$$= 0\left(\frac{3}{9}\right) + 2\left(\frac{4}{9}\right) + 4\left(\frac{2}{9}\right)$$

$$\mu_Y = E[Y] = \frac{16}{9} = 0.17$$

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a) (4 points) Calculate and plot the PMF of X

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c) (4 points) Calculate Mean of Y.

d) (4 points) Calculate Variance of Y

(d)

$$\text{var}(Y) = E[Y^2] - (\mu_Y)^2$$

$$E[Y^2] = \sum_y y^2 p_Y(y) \quad \text{(Expected Value Rule)}$$

$$= 0^2\left(\frac{3}{9}\right) + 2^2\left(\frac{4}{9}\right) + 4^2\left(\frac{2}{9}\right)$$

$$= \frac{16}{9} + \frac{32}{9}$$

$$E[Y^2] = \frac{48}{9}$$

$$\text{var}(Y) = \frac{48}{9} - \left(\frac{16}{9}\right)^2$$

$$\text{var}(Y) = 2.17$$

Question 4 - (4 points)

Consider two random variables X & Y:

X = Temperature in Karachi on Oct 29, 2021 in Celsius (°C)

Y = Temperature in Karachi on Oct 29, 2021in Fahrenheit (°F)

Assume that you have been provided the PMF of X and the corresponding PMF of Y.

Case 1: You take the mean of X and then convert this mean value to °F

Case 2: You take the mean of Y

Will you get the same result in Case 1 and Case 2? Why or why not? Justify your answer.

(Hint: Celsius to Fahrenheit Conversion Formula: $°\mathrm{F} = \mathbf{1.8}°\mathrm{C} + \mathbf{32}$ ***)***

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(Hint: Celsius to Fahrenheit Conversion Formula: °F = $\mathbf{1.8}$°C + $\mathbf{32}$)

If Y is a linear function of X, i.e.

$$Y = aX + b$$

$$E[Y] = aE[X] + b$$

Here,

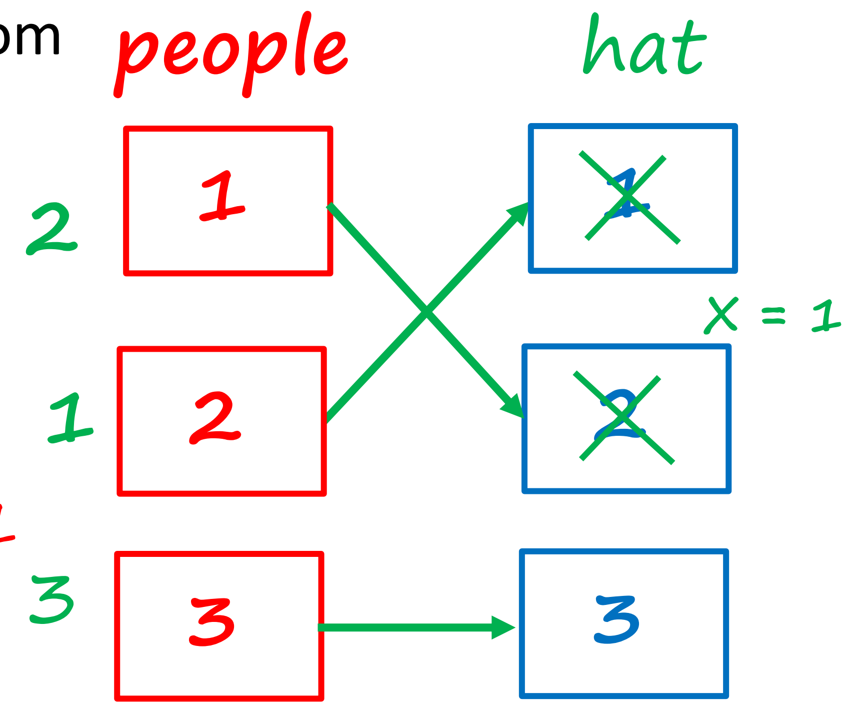
$$Y = 1.8X + 32$$

$$\Rightarrow \underbrace{E[Y]}_{\text{Case 2}} = \underbrace{1.8E[X] + 32}_{\text{Computation in Case 1}}$$

Therefore, you will get the same result in both cases.

End of Chapter Exercise - The hat problem

- *n* people throw their hats in a box and then pick one at random
  - ❖ All permutations equally likely $1/n!$
  - ❖ Equivalent to picking one hat at a time



- X: number of people who get their own hat
  - ❖ Find $\boxed{\mathbf{E}[X]}$ $= E[X_1] + \ldots + E[X_n] = n * 1/n = 1$

$$X_i = \begin{cases} 1, & \text{if i selects own hat} \\ 0, & \text{otherwise} \end{cases}$$

$$\boldsymbol{X = X_1 + \ldots + X_n}$$

❖ $\mathbf{E}[X_i] = E[X_1] = P(X_1 = 1) = 1/n$

$$\frac{1}{3} \cdot \frac{1}{2} \cdot 1 = \frac{1}{3!}$$

$P_X(k)$: hard!

$$\sum_k k\, p_X(x)$$

The variance in the hat problem

$n = 2$

$X_1 = 1 \Rightarrow X_2 = 1$

$X_1 = 0 \Rightarrow X_2 = 0$

- X: number of people who get their own hat
  - ❖ Find var(X)

$$X_i = \begin{cases} 1, & \text{if i selects own hat} \\ 0, & \text{otherwise} \end{cases}$$

$$X = X_1 + \dots + X_n$$

$$X^2 = \overbrace{\sum_i X_i^2}^{n} + \overbrace{\sum_{i,j:\, i \neq j} X_i X_j}^{n(n-1)}$$

- $\boxed{var(X)} = E[X^2] - (E[X])^2 = 2 - 1 = \boxed{1}$

- $E[X_i^2] = E[X_1^2] = E[X_1] = 1/n$

$$\Rightarrow E[X^2] = n\frac{1}{n} + n(n-1)\frac{1}{n} * \frac{1}{n-1}$$

- for $i \neq j$: $E[X_i X_j] = E[X_1 X_2] = P(X_1 X_2 = 1) = P(X_1 = 1, X_2 = 1)$

$$= P(X_1 = 1) . P(X_2 = 1 \mid X_1 = 1) = \frac{1}{n} * \frac{1}{n-1}$$